

Integrals in Context: Average Value, Accumulation, and Motion

Answer Key

AP Calculus AB/BC

Quick Answers

- | | |
|--|--|
| 1. the average value = 16 | 7. the average temperature = $18 + \frac{12}{\pi}$ |
| 2. the litres added = 84 L | 8. the litres lost = $200 - \frac{200}{e}$ L |
| 3. the displacement = 16 m | 9. the turning time = 4, the position at $t = 6 = 9$ m |
| 4. the average value = $\frac{2}{\pi}$ | 10. the total distance = $\frac{28}{3}$ m, the average velocity = $\frac{4}{3}$ m/s, — |
| 5. the kilograms added = 40.5 kg | |
| 6. the times the particle turns = 1, 3, the total distance = 4 m | |
-

What is verified

13 of 14 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus AB/BC

FUN-6 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 14 tagged checks.

1. Find the average value of $f(x) = 3x^2$ on $[0, 4]$.

Solution: The average value is $\frac{1}{b-a} \int_a^b f$, here $\frac{1}{4} \int_0^4 3x^2 dx$:

$$\frac{1}{4} \int_0^4 3x^2 dx = \frac{1}{4} [x^3]_0^4 = \frac{64}{4}.$$

So the average value is

$f_{\text{avg}} = 16$

2. Water enters a tank at $r(t) = 4t + 2$ litres per minute. How much enters during the first 6 minutes?

Solution: Accumulated change is the integral of the rate:

$$\int_0^6 (4t + 2) dt = \left[2t^2 + 2t \right]_0^6 = 72 + 12.$$

Litres per minute times minutes gives litres.

84 L

- 3.** A particle has velocity $v(t) = 3t^2 - 12$ m/s. Find its displacement on $[0, 4]$.

Solution: Displacement is the integral of velocity (no absolute value — the question asks for net change of position):

$$\int_0^4 (3t^2 - 12) dt = \left[t^3 - 12t \right]_0^4 = 64 - 48.$$

So the displacement is

16 m

- 4.** Find the average value of $g(x) = \sin x$ on $[0, \pi]$.

Solution: $\frac{1}{\pi} \int_0^\pi \sin x dx = \frac{1}{\pi} \left[-\cos x \right]_0^\pi = \frac{1}{\pi} (1 + 1)$, so the average value is

$g_{\text{avg}} = \frac{2}{\pi} \approx 0.637$

- 5.** Sand accumulates at $r(t) = 9 - t$ kilograms per minute for $0 \leq t \leq 9$. How much is added?

Solution:

$$\int_0^9 (9 - t) dt = \left[9t - \frac{t^2}{2} \right]_0^9 = 81 - \frac{81}{2}.$$

So the pile gains

40.5 kg

- 6.** $v(t) = t^2 - 4t + 3$ m/s on $[0, 4]$. (a) When does the particle turn? (b) Total distance travelled?

Solution (a): The particle turns where the velocity changes sign, so solve $t^2 - 4t + 3 = (t - 1)(t - 3) = 0$.

$t = 1, t = 3$

Solution (b): Total distance is $\int_0^4 |v(t)| dt$, split at the turning times. On $[0, 1]$ and $[3, 4]$ the velocity is positive; on $[1, 3]$ it is negative. With $F(t) = \frac{t^3}{3} - 2t^2 + 3t$,

$$\int_0^4 |v| = |F(1) - F(0)| + |F(3) - F(1)| + |F(4) - F(3)| = \frac{4}{3} + \frac{4}{3} + \frac{4}{3}.$$

So the total distance is

4 m

- 7.** $T(t) = 18 + 6 \sin\left(\frac{\pi t}{12}\right)$ degrees Celsius. Find the average temperature over $[0, 12]$.

Solution:

$$\frac{1}{12} \int_0^{12} \left(18 + 6 \sin \frac{\pi t}{12} \right) dt = \frac{1}{12} \left[18t - \frac{72}{\pi} \cos \frac{\pi t}{12} \right]_0^{12} = \frac{1}{12} \left(216 + \frac{144}{\pi} \right).$$

So the average temperature is

$$T_{\text{avg}} = 18 + \frac{12}{\pi} \approx 21.820$$

8. Water leaks at $r(t) = 20e^{-0.1t}$ litres per hour. How much leaks out during the first 10 hours?

Solution:

$$\int_0^{10} 20e^{-0.1t} dt = \left[-200e^{-0.1t} \right]_0^{10} = -200e^{-1} + 200.$$

So the tank loses

$$200(1 - e^{-1}) \approx 126.42 \text{ L}$$

9. $v(t) = 4 - t$ m/s, $x(0) = 3$ m. (a) When does the particle turn? (b) Where is it at $t = 6$?

Solution (a): Solve $4 - t = 0$.

$$t = 4$$

Solution (b): Position is the starting value plus the accumulated displacement, so

$$x(t) = 3 + \int_0^t (4 - u) du = 3 + 4t - \frac{t^2}{2}. \text{ At } t = 6 \text{ this is } 3 + 24 - 18.$$

$$s(6) = 9 \text{ m}$$

Note that the particle has travelled further than 6 metres — it moved right until $t = 4$ and then back — which is why part (b) asks for position, not distance.

10. $v(t) = t^2 - 6t + 8$ m/s on $[0, 5]$, starting at $x = 1$ m. (a) Total distance. (b) Average velocity. (c) Why is the average velocity not the average of the extreme velocities?

Solution (a): $v = (t - 2)(t - 4)$, so the sign changes at $t = 2$ and $t = 4$. With $F(t) = \frac{t^3}{3} - 3t^2 + 8t$, the three pieces measure $\frac{20}{3}$, $\frac{4}{3}$ and $\frac{4}{3}$, so

$$\int_0^5 |v| dt = \frac{20}{3} + \frac{4}{3} + \frac{4}{3}.$$

The total distance is

$$\frac{28}{3} \approx 9.33 \text{ m}$$

Solution (b): Average velocity is the average value of v , i.e. displacement over elapsed time:

$$\frac{1}{5} \int_0^5 (t^2 - 6t + 8) dt = \frac{1}{5} \cdot \frac{20}{3}.$$

So the average velocity is

$$\frac{4}{3} \approx 1.33 \text{ m/s}$$

Solution (c) — instructor-judged. A correct response says the average value divides the integral by the length of the interval, which weights every instant equally; averaging only the largest and smallest velocities uses two instants and ignores how long the particle spends near each speed. Full credit contrasts the two weightings.